NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH- POUND

MIL-PRF-82815(OS)
29 December 1989
SUPERSEDING
WS 22376
25 October 1983

PERFORMANCE SPECIFICATION

LIQUID, ANTISTATIC

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a topical antistatic liquid, referred to herein as the antistatic liquid.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-1752	Cushioning Material, Packaging, (Unicellular Polyethylene Foam, Flexible)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-101 Test Procedures for Packaging Materials

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from Military Specifications and Standards, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR 43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR 43 are available from the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1875 Density of Adhesives in Fluid Form (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the antistatic liquid shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 **Properties.** Physical and chemical properties of the antistatic liquid shall be in accordance with table I.

3.3 **Safety.**

3.3.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3.2 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

TABLE I. Physical and chemical properties.

Property	Minimum	Maximum
Density, lb/gal	8.17	8.57
Static decay, seconds	---	2.00
Ionic content (percent wt/wt)		
Fluoride	---	.001
Chloride	---	.001
Sulfate	---	.001
Phosphate	---	.001

3.4 **Shelf life.** The antistatic liquid shall have an unlimited shelf life when stored at 80 ± 20°F.

3.5 **Workmanship.** The workmanship shall be such that the antistatic liquid is free from contamination and defects that could render it unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspection (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. Quality conformance inspections.

Tests and Examinations	Requirement	Test Method
Visual examination	3.5 and Section 5	4.5.1
Density	3.2	4.5.2
Static decay	3.2	4.5.3
Ionic content1/	3.2	4.5.4

1/ Unless otherwise specified, contractor certification verifying conformance to this requirement shall be required.

4.4.1 Lot. A lot shall consist of all antistatic liquid subject to inspection at the same time from one of the following:

a. A single batch (see 6.3.1)

b. A continuous process (see 6.3.2)

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.2 Sampling. Sampling shall be in accordance with table III.

4.4.3 Classification of characteristics. The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

(M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

TABLE III. Sampling.

Number of Containers in Lot	Number of Containers to be Selected
1 - 3	All
4 - 15	3
16 - 25	4
26 - 40	5
41 - 65	7
66 - 110	10
111 - 180	15

4.5 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.5 and section 5.

4.5.2 **Density.** Density (lb/gal) shall be determined in accordance with ASTM D 1875.

4.5.3 **Static decay (M101).** Static decay shall be determined in accordance with FED-STD-101, Method 40461, to 500 volts at 15 percent relative humidity. The specimens used for the test shall be 5 x 3 x 1 inch blocks of polyethylene foam conforming to PPP-C-1752, Type V, Class 2 and Type III, Class 2, with the antistatic liquid applied by spray or dip process.

4.5.4 **Ionic content.** Ionic content shall be determined using ion chromatography.

4.6 **Inspection of packaging.** The antistatic liquid containers and container markings shall be visually examined to verify compliance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The antistatic liquid shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.4. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.
b. Manufacturer's name and address.
c. Date of manufacture.
d. Shelf life expiration date.
e. Storage conditions.
f. Toxicity warning (if applicable).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The antistatic liquid is intended for use in the manufacture of rocket motor shipping containers.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Whether first article inspection is required (see 3.1 and 6.4)

d. Material Safety Data Sheets Required (see 3.3.2 and 6.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

e. DOD FAR 52.246-15 (see table II)

f. Special marking, if other than as specified in 5.2.2

g. Safety precautions required (see 6.7)

6.3 Definitions.

6.3.1 Batch. A batch is that amount of antistatic liquid made as one unit in the manufacturing process.

6.3.2 Continuous process. A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the antistatic liquid is accumulated for the lot.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Possible material. The following material has been found to meet the requirements of this specification. This is given for information only and is not restrictive. RESQUE 2 manufactured by DM Laboratories, Inc., Fullerton, California, has been found to meet the requirements of this specification.

6.6 Material Safety Data Sheets. MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102. NOTE: When this specification

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.8 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N802)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82815	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
ANTISTATIC LIQUID

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2020-10-28 10:07:21Z
Check the source to verify that this is the current version before use.